



The worm with three sexes

In the eastern reaches of the Sierra Nevada desert, researchers have discovered a worm that has three sexes. <https://digit.in/nov19-01>



The paradox of reproduction

This paper attempts to shed light on the mystery of the origin of sexual reproduction using sociobiology. <https://digit.in/nov19-02>

THE BIRDS AND THE BEES

One of the long standing mysteries in biology, is the origin of sexual reproduction.

Aditya Madanapalle |
aditya@digit.in

99.99 percent of eukaryotes reproduce sexually. The earliest

life forms on Earth reproduced asexually, so it follows that sexual reproduction evolved and persisted. Exactly why though, is an unsolved problem. Theoretically, species pay a steep price for sexual reproduction. Only about 50 percent of the members of a species can produce offspring, as compared to species that reproduce asexually. A species that reproduces sexually can only grow at half the rate of a species that reproduces asexually. Additionally, the creatures have to find a mate, which can be demanding in terms of energy. There is also a heritability cost. In most cases, a parent can only pass on 50 percent of their genes to the next generation. Genes may evolve aggressive tactics, where they favour their own transmission over others, with no clear advantage to the resulting organism. To overcome all these disadvantages, the benefits of sexual reproduction must be

compelling indeed. One of the biggest advantages of sexual reproduction is genetic variability. As the genes from two parents are recombined, there remains a healthy trial and error method, which allows a species to adapt better to its environment over time. Another advantage is that, in case there are mutations or degradations in the DNA, the healthy versions will be transferred to the next generation. However, sexual reproduction does not always produce increased variability. For example, think of two species, A and B. Both species have individuals with a range of sizes, from small to large. A reproduce sexually, while B reproduce asexually. The variation in height is conserved in B, but as the generations go by, all the organisms in species A will tend to be of medium size.

Stepping back from the debate on whether or not sexual reproduction is an advantage, is the question of why it originated in the first place. Any conjectures on the matter are extremely difficult to verify experimentally, although evolutionary computation is potentially a viable approach. The technique allows a rapid simulation of a large number of generations, to check if the same

kinds of behaviour and traits emerge in organisms, given a set of initial conditions and external pressures. Sexual reproduction is thought to have emerged about 2 billion years ago, from the last eukaryotic common ancestor (LECA). Scientists suspect that sexual reproduction may have originated several times, but one of those times, it persisted. Most of the research is focused on the maintenance of sexual reproduction, but not its origin. There are some snails, insects, crustaceans and sharks where the species have been unable to maintain sexual reproduction. These species have only female members, and they give birth through a form of asexual reproduction known as parthenogenesis, which is the Greek word for virgin birth. Facultative parthenogenesis is extremely rare, where the female can produce offspring either sexually or asexually. This form of reproduction is most commonly seen in mayflies. Some Komodo dragons, snakes and sharks have been known to bear offspring when isolated from males in captivity, even though the species do not normally produce progeny in that manner.